

Use Case Name	Create Registration	
Use Case ID	UC01	
Level	Detailed	
Primary Actors	Client (Supervisor, Creator, Editor)	
Goal	The user provides details to create an account and get access to the system.	
Precondition	1. The registration page is displayed. 2. The client does not exist.	
Failure Condition	1. The user provides invalid parameters. 2. System returns a “time out” error after 60 seconds.	
Success Condition	1. Create an account successfully. 2. The user can log in to the system.	
Trigger	The user clicks the register button on the login page.	
Basic Flow	Actor Action	System response
	1.The client provides details	2.Check if the client exists. If registration has existed, go to 2.1. Else, go to 3.
		3.Create a registration with the client details if details are complete and meet the requirement. Else if part of details is empty, go to 3.1. Else if part of details does not meet the requirements, go to 3.2.
Abnormal Events Flow	Actor Action	System response
	2.1The registration has existed.	2.1 Display a message indicating that the user exists.
	3.1The data provided is empty.	3.1Display a message indicating that part of the details is empty.
	3.2The provided data does not meet the format requirements.	3.2Display a message indicating that some data does not meet the requirements.
Alternative Flow		

Use Case Name	Login	
Use Case ID	UC02	
Level	Detailed	
Primary Actors	Client (Supervisor, Creator, Editor)	
Goal	A process of entering valid identification information, including a username and a password, to access the system. It verifies the user's identity and grants them access to their account or authorized resources.	
Precondition	1. The login page is displayed. 2. The client exists.	
Failure Condition	1. Username or password is invalid. 2. System returns a “time out” error after 60 seconds.	
Success Condition	1. The user logs in successfully.	
Trigger	The user clicks the login button.	
Basic Flow	Actor Action	System response
	1.The client provides username and password.	2.Check if the client does not exist. If the client does not exist, go to 2.1. Else go to 3.
		3.The system searches the user in the database if the username and the password are correct, and displays the index. Else if the username or password is empty, go to 3.1. Else if the username and password are not correct, go to 3.2.
Abnormal Events Flow	Actor Action	System response
	2.1The registration has not existed.	2.1 Display a message indicating that the user has not existed.
	3.1The username or password is empty.	3.1Display a message indicating that the User Name or Password is empty.
	3.2The username and password are not correct.	3.2Display a message indicating that the username and password are not correct. .
Alternative Flow		

Use Case Name	Create Illustration	
Use Case ID	UC03	
Level	Detailed	
Primary Actors	Novel Writer, Social Media User	
Goal	Users input text to submit the request, the system analyzes the meaning of the text through the language model, the system inputs the text into the image generation model and returns the illustrations.	
Precondition	1. The actor has already logged into the system. 2. The actor is authorized to use this feature.	
Failure Condition	1. Created illustration is not qualified. 2. System returns a “time out” error after 60 seconds.	
Success Condition	1. The system returns the illustration successfully. 2. At the same time, the illustration is very close and correctly expresses the text content.	
Trigger	The user clicks the create illustration button on the image creation page.	
Basic Flow	Actor Action	System response
	1.The actor provides the text.	2.The system receives the text then refines and simplifies it using a large language model.
		3.The system input the summary to the image generation model to create the illustration.
	4.If the actor is not satisfied can be regenerated from the selection. Then go to 2.	
Abnormal Events Flow	Actor Action	System response
	2.1The text is empty.	2.1Display a message indicating that the text is empty.
Alternative Flow	1. If the user is not satisfied, the illustration can be modified according to the tool section provided by the system. 2.If the illustration generation is for the novel, the characters generated by the system should be the same, and the illustration style should be similar.	

Use Case Name	Create Poster	
Use Case ID	UC04	
Level	Detailed	
Primary Actors	Post Designer	
Goal	The user enters the type of product and the text description of the poster he wants and uploads the image of the corresponding product. The system uploads the received text description to the language large model for refining and inputs the refined text and product image together into the image large model and returns the poster.	
Precondition	1. The actor has already logged into the system. 2. The actor is authorized to use this feature.	
Failure Condition	1. Created poster is not qualified. 2. System returns a “time out” error after 60 seconds.	
Success Condition	1. The system returns the poster successfully. 2. At the same time, the poster is very close and correctly expresses the text content.	
Trigger	The user clicks the create poster button on the image creation page.	
Basic Flow	Actor Action	System response
	1.The actor provides the text description.	
	2.the actor uploads the image of the product to the system.	3.The system receives the text then refines and simplifies it using a large language model.
		4.The system inputs the summary and the product image to the image generation model to create the poster.
	5.If the actor is not satisfied can be regenerated from the selection. Then go to 3.	
Abnormal Events Flow	Actor Action	System response
	2.1The text description is empty.	2.1Display a message indicating that the text is empty.
Alternative Flow	1. If the user is not satisfied, the illustration can be modified according to the tool section provided by the system.	

Use Case Name	Create Icon	
Use Case ID	UC05	
Level	Detailed	
Primary Actors	Social media user	
Goal	Users enter the text description of the icon they want, and can also choose to upload a picture. If the user does not upload a picture, the icon is generated based on the text description. If a picture is uploaded, an icon is generated based on the picture and text description.	
Precondition	1. The actor has already logged into the system. 2. The actor is authorized to use this feature.	
Failure Condition	1. Created icon is not qualified. 2. System returns a “time out” error after 60 seconds.	
Success Condition	1. The system returns the icon successfully. 2. At the same time, the icon is very close and correctly expresses the text content. If the icon is based on the picture, the icon can retain the original image features.	
Trigger	The user clicks the create icon button on the image creation page.	
Basic Flow	Actor Action	System response
	1.The actor provides the text description.	
	2.the actor uploads the picture to the system.(optional)	3.The system receives the text then refines and simplifies it using a large language model.
		4.The system inputs the summary and the product image(if have) to the image generation model to create the poster.
Abnormal Events Flow	Actor Action	System response
	2.1The text description is empty.	2.1Display a message indicating that the text is empty.
Alternative Flow	1.If the user is not satisfied, the illustration can be modified according to the tool section provided by the system.	

Use Case Name	Create Social Media Post	
Use Case ID	UC06	
Level	Detailed	
Primary Actors	Social Media User	
Goal	Social media users can input some prompts to generate pictures based on their needs. They can also use system models to optimize their post content. Users also can modify the content after the generation and choose the way of social media they want to share with.	
Precondition	1. The actor has already logged into the system. 2. The actor is authorized to use this feature.	
Failure Condition	1. Created context or pictures are not qualified. 2. System returns a “time out” error after 60 seconds.	
Success Condition	1. The system returns the picture based on users’ needs successfully. 2. The system returns the content of social media posts successfully. 3. The users can modify the content. 4. The users can choose the way to share the generated post.	
Trigger	The user clicks the create poster button on the context creation page.	
Basic Flow	Actor Action	System response
	1.The actor provides the text description.	
	2.The actor inputs the prompt of the picture they want to generate to the system.(optional)	3.The system receives the text, refines and simplifies the text using a large language model.
		4.The system inputs the summary and the product image(if have) to the image generation model to create the poster.
Abnormal Events Flow	Actor Action	System response
	1.1The actor makes no modification of content.	1.1The system has no response to this action or shows a message of error.
	2.1 The actor inputs invalid prompt.	2.1The system fails to generate the picture and shows a message of error.
Alternative Flow	1.The Actor chooses one type of social media used to share the outcome generated.	

Use Case Name	Content Audit	
Use Case ID	UC07	
Level	Detailed	
Primary Actors	Supervisor	
Goal	Supervisor can audit the archive created by users and give users feedback respectively.	
Precondition	- Supervisor has an available account and logs in successfully. - Supervisor clicks the “Content Audit” button and jumps to the audit page successfully.	
Failure Condition	- Supervisor cannot access the record lists. - Supervisor has no privilege of the audit. - Supervisor cannot make any comments and operations.	
Success Condition	Supervisor can view the archive online and give the feedback successfully.	
Trigger	- Supervisor logs in the system. - Click the audit button.	
Basic Flow	Actor Action	System response
	1.The actor chooses an archive and views it online.	2. The system returns the chosen archive to the actor.
	3. The actor submits the feedback for this archive.	4. The system receives the feedback text and returns a sign of success to the actor.
		5. The system uses the feedback text to create the audit archive.
Abnormal Events Flow	Actor Action	System response
	1.1 The actor cannot view the archive.	1.1 The system returns a sign of failure to the actor and displays a message of “Fail to load the archive”.
	2.1 The feedback text is empty.	2.1 The system returns a sign of failure to the actor and displays a message of “The feedback text is empty”.
Alternative Flow		

Use Case Name	AI Model Update	
Use Case ID	UC08	
Level	Detailed	
Primary Actors	Administrator	
Goal	Administrators can update the AI generative model provided to users.	
Precondition	- Administrators can log in their account successfully. - Administrator accounts are setting up permissions correctly. - Administrators can access the “AI Model Update” page successfully. - The system is in maintenance condition.	
Failure Condition	- The AI model is not updated. - The AI model cannot be uploaded to the system.	
Success Condition	1.The system successfully updates its current AI model and returns a sign of success to the administrator. 2. A system modification history is created and logged.	
Trigger	- The administrator logs in to the system. - The administrator clicks the “update model” button on the setting page.	
Basic Flow	Actor Action	System response
	1. The actor logs in the administrator account and access the system update page.	
	2. The actor uploads the latest AI model to the system.	
	3. The actor input the password of the account again for identity check.	4. The system checks the password and returns a sign of certification success.
		5. The system checks the availability of the AI model uploaded by the actor.
		6. The system changes the AI model to the latest one, returns a sign of success and creates a system modification history.
Abnormal Events Flow	Actor Action	System response
	2.1 The model was incomplete.	2.1 The upload failed and the system will notify the administrator with an error message.
Alternative Flow		

Use Case Name	Context Edit	
Use Case ID	UC09	
Level	Detailed	
Primary Actors	Users	
Goal	Users can edit the generated context and alter the font and font size of the context.	
Precondition	1. The user logs in the system successfully. 2. The system generated qualified context.	
Failure Condition	1. The user cannot log in to the system. 2. The system generates unqualified context. 3. The system returns a “time out” error after 60 seconds.	
Success Condition	1. The user can edit the generated context. 2. The user can alter the font and font size of the context.	
Trigger	1. The user logs into the system. 2. The user clicks the context generation button. 3. After getting the result, the user clicks the edit button.	
Basic Flow	Actor Action	System response
	1. The user logs into the system.	
	2. The user clicks the context generation button.	
		3. The system returns the result.
	4. The user clicks the edit button.	
Abnormal Events Flow	Actor Action	System response
	2.1 The user does not log in the system.	2.1 The system returns a sign of failure.
		2.2 The system fails to generate context.
	2.3 The user does not click the edit button.	
		2.4 The system has no response when a user clicks edit button or context cannot be edited.
Alternative Flow		

Use Case Name	Forwarding	
Use Case ID	UC10	
Level	Detailed	
Primary Actors	Supervisor, Creator	
Goal	Creators can forward the results after the content creation. Supervisors can forward the results after the content audit.	
Precondition	1. The user logs in the system successfully. 2. The user created the context or audited the context successfully.	
Failure Condition	1. The user cannot log in to the system. 2. The context creation or the context audit is not successful. 3. The system returns a “time out” error after 60 seconds.	
Success Condition	1. The system forwards the context creation or audit successfully.	
Trigger	1. The user logs into the system. 2. The user creates the context creation or audit. 3. The user clicks the forwarding button on the context page or on the history page.	
Basic Flow	Actor Action	System response
	1. The user logs into the system.	
	2. The user clicks the context generation button or the history button.	
		3. The system returns the result.
	4. The user clicks the forwarding button.	
Abnormal Events Flow	Actor Action	System response
	1.1 The user does not log in the system	1.1 The system returns a sign of failure.
	4.1 The user chooses an incomplete context.	4.1 The system fails to forward context.
Alternative Flow		